

Fundamentals Physics

Twelfth Edition

Halliday

Chapter 16

Waves - I

Section 16.1 Transverse Waves

Transverse Waves Learning Objectives #1 - #3

1. Identify the three main types of waves.
2. Distinguish between transverse and longitudinal waves.
3. Given a displacement function for a transverse wave, determine amplitude y_m , angular wave number k , angular frequency ω , phase constant ϕ , and direction of travel; and calculate the phase $kx \pm \omega t + \phi$ and the displacement at any given time and position.

Transverse Waves Learning Objectives #4 - #6

4. Given a displacement function for a transverse wave, calculate the time between two given displacements.
5. Sketch a graph of a transverse wave as a function of position, identifying amplitude y_m and wavelength λ where the slope is greatest, where it is zero, and where the string elements have positive velocity, negative velocity, and zero velocity.
6. Given a graph of displacement versus time for a transverse wave, determine amplitude y_m and period T .

Transverse Waves Learning Objectives #7 - #9

7. Describe the effect on a transverse wave of changing phase constant ϕ .
8. Apply the relation between the wave speed v , the distance traveled by the wave, and the time required for that travel.
9. Apply the relationships between wave speed v , angular frequency ω , angular wave number k , wavelength λ , period T , and frequency f .

Transverse Waves Learning Objectives #10 - #13

10. Describe the motion of a string element as a transverse wave moves through its location, and identify when its transverse speed is zero and when it is maximum.
11. Calculate the transverse velocity $u(t)$ of a string element as a transverse wave moves through its location.
12. Calculate the transverse acceleration $a(t)$ of a string element as a transverse wave moves through its location.
13. Given a graph of displacement, transverse velocity, or transverse acceleration, determine the phase constant ϕ .

Types of Waves

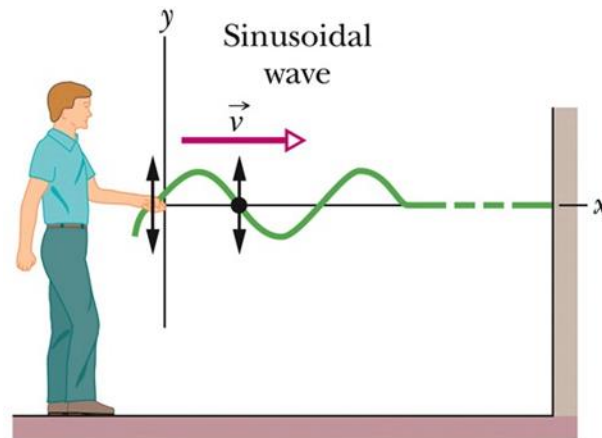
- 1. Mechanical Waves:** They are governed by Newton's laws, and they can exist only within a material medium, such as water, air, and rock. Examples: water waves, sound waves, and seismic waves.
- 2. Electromagnetic waves:** These waves require no material medium to exist. Light waves from stars, for example, travel through the vacuum of space to reach us. All electromagnetic waves travel through a vacuum at the same speed $c = 299\,792\,458\text{ m/s}$.
- 3. Matter waves:** These waves are associated with electrons, protons, and other fundamental particles, and even atoms and molecules. Because we commonly think of these particles as constituting matter, such waves are called matter waves.

Gravitational Waves

- In 1916, Albert Einstein predicted that when any mass accelerates, it sends out *gravitational waves* that are oscillations of space itself (more precisely, spacetime).
- In normal circumstances, the oscillations are so small as to be undetectable. The first direct detection of the waves came in 2015 when a detector based on the design of Rainer Weiss of MIT recorded the waves due to the merger of two distant black holes.
- The oscillations were much less than the radius of a proton.

Example of Transverse Wave

A sinusoidal wave is sent along the string. A typical string element moves up and down continuously as the wave passes. This is **transverse wave**. The element's motion is perpendicular to the wave's direction of travel, so the pulse is a *transverse wave*.



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Figure 16.1.1 (a)

Example of Longitudinal Wave

A sound wave is set up in an air- filled pipe by moving a piston back and forth. Because the oscillations of an element of the air (represented by the dot) are parallel to the direction in which the wave travels, the wave is a **longitudinal wave**.

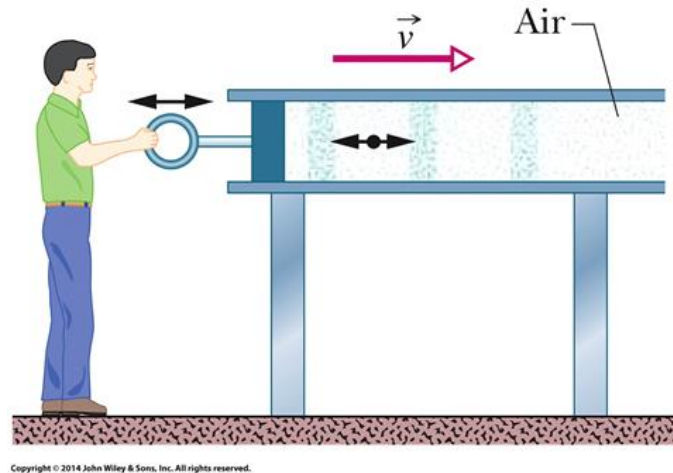


Figure 16.1.1 (b)

Sinusoidal Wave

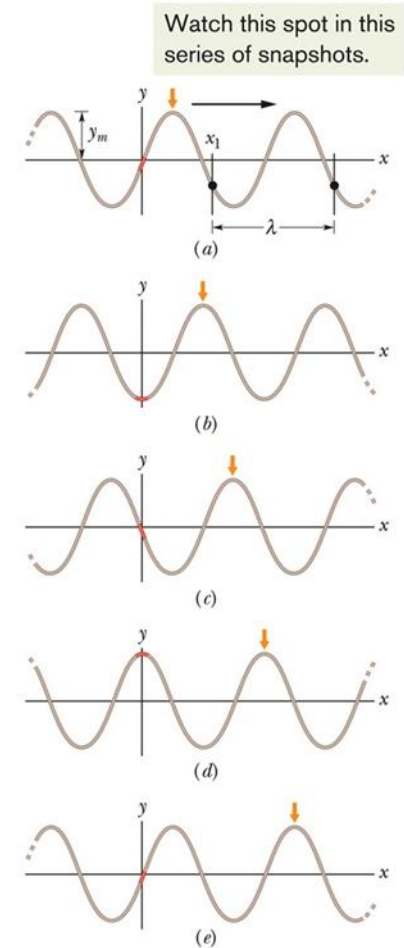
Five “snapshots” (y vs x each at a constant time) of a string wave traveling in the positive direction along an x axis. The amplitude y_m is indicated. A typical wavelength λ , measured from an arbitrary position x_1 , is also indicated.

Amplitude
Displacement
Angular wave number
Position
Oscillating term
Phase
Time
Angular frequency

$$y(x,t) = y_m \sin(kx - \omega t)$$

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The sine function describes the shape of the wave



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Figure 16.1.4

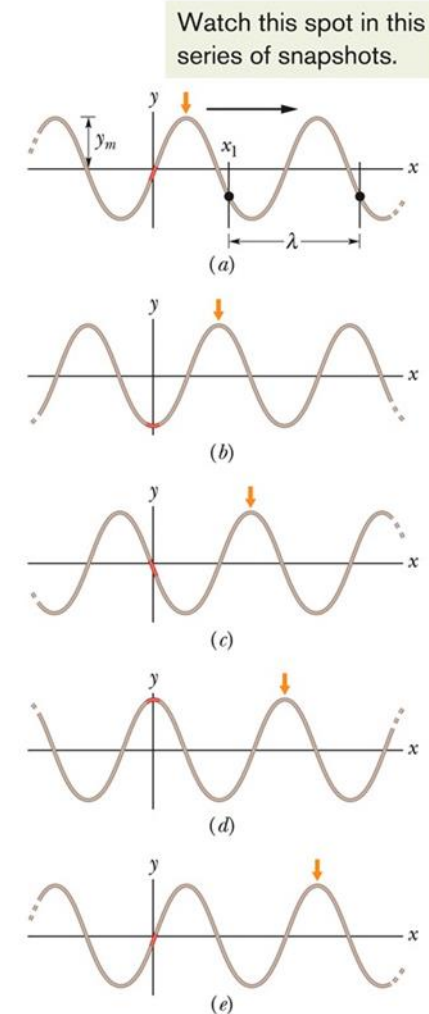
Wave Parameters

Period, Wave Number, Angular Frequency and Frequency

$$k = \frac{2\pi}{\lambda} \quad (\text{angular wave number}). \quad \text{Equation 16.1.5}$$

$$\omega = \frac{2\pi}{T} \quad (\text{angular frequency}). \quad \text{Equation 16.1.8}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi} \quad (\text{frequency}). \quad \text{Equation 16.1.9}$$



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Figure 16.1.4

Wave Speed

The Speed of a Traveling Wave

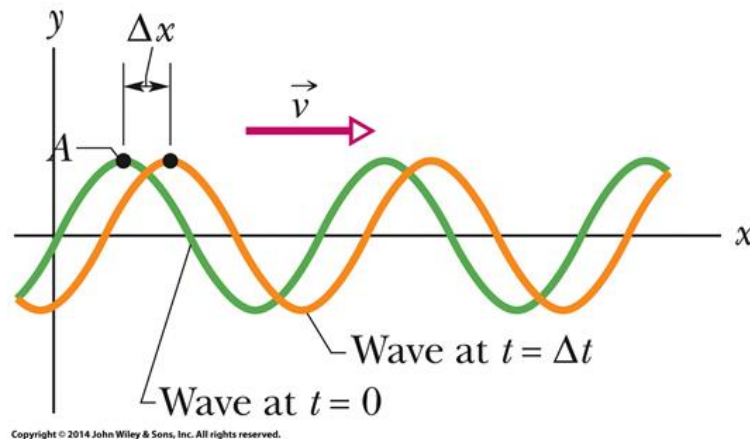


Figure 16.1.7

Two snapshots of the wave: at time $t = 0$, and then at time $t = \Delta t$. As the wave moves to the right at velocity v , the entire curve shifts a distance Δx during Δt .

Wave Speed Definition

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f \quad (\text{wave speed}). \quad \text{Equation 16.1.13}$$

$$y(x, t) = y_m \sin(kx + \omega t). \quad \text{Equation 16.1.15}$$

Section 16.2 Wave Speed on a Stretched String

Wave Speed on a Stretched String

Learning Objectives

- Calculate the linear density μ of a uniform string in terms of the total mass and total length.
- Apply the relationship between wave speed v , tension τ , and linear density μ .

Linear Density Relationship to Wave Speed for Stretched String

The speed of a wave on a stretched string is set by properties of the string (i.e. linear density), not properties of the wave such as frequency or amplitude. Tau is the tension (in N) in the string.

$$\mu = \frac{m}{l} \quad (\text{linear density})$$

$$v = \sqrt{\frac{\tau}{\mu}} \quad \leftarrow (\text{speed}),$$

Equation 16.2.5

Section 16.3 Energy & Power of a Wave Traveling along a String

Energy and Power of a Wave Traveling along a String Learning Objective

- Calculate the average rate at which energy is transported by a transverse wave.

Kinetic Energy Transmission of a Wave Traveling along a String

- When we set up a wave on a stretched string, we provide energy for the motion of the string. As the wave moves away from us, it transports that energy as both kinetic energy and elastic potential energy.
- **The Rate of Energy Transmission** The kinetic energy dK associated with a string element of mass dm is given by

$$dK = \frac{1}{2} dm u^2, \quad \text{Equation 16.3.1}$$

where u is the transverse speed of the oscillating string element

Average Power of a Wave Traveling along a String

- The **average power** of, or average rate at which energy is transmitted by, a sinusoidal wave on a stretched string is given by

where: μ = linear density, ω = angular frequency, v = wave speed, $y_{\max}$ = amplitude

$$P_{\text{avg}} = \frac{1}{2} \mu v \omega^2 y_m^2. \quad \text{Equation 16.3.7}$$

The factors μ and v in this equation depend on the material and tension of the string. The factors ω and y_m depend on the process that generates the wave.

Section 16.4 The Wave Equation

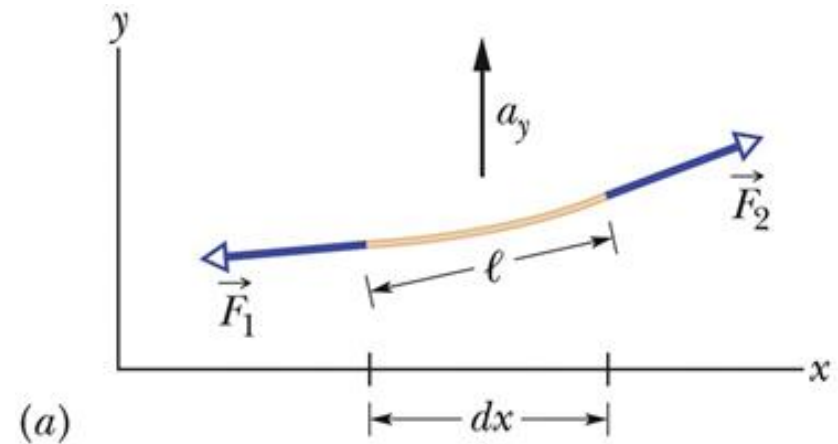
The Wave Equation Learning Objective

- For the equation giving a string-element displacement as a function of position x and time t , apply the relationship between the second derivative with respect to x and the second derivative with respect to t .

Newton's Second Law or a Wave Traveling on a Stretched String

By applying Newton's second law to the element's motion, we can derive a general differential equation, called the wave equation, that governs the travel of waves of any type.

- a) A string element as a sinusoidal transverse wave travels on a stretched string. Forces $\vec{F}_1$ and $\vec{F}_2$ act at the left and right ends, producing acceleration with a vertical component a_y .

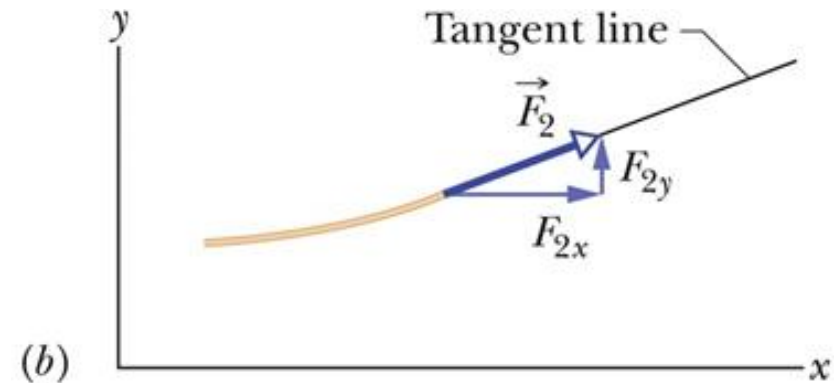


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Figure 16.4.1 (a)

Force Resolution for Stretched String Traveling Wave

- b) The force at the element's right end is directed along a tangent to the element's right side.



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Figure 16.4.1 (b)

The Wave Equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \quad (\text{wave equation}). \quad \text{Equation 16.4.12}$$

This is the general differential equation that governs the travel of waves of all types. Here the waves travel along an x axis and oscillate parallel to the y axis, and they move with speed v , in either the positive x direction or the negative x direction.

Section 16.5 Interference of Waves

Wave Interference Learning Objectives

- Apply the principle of superposition to show that two overlapping waves add algebraically to give a resultant (or net) wave.
- For two transverse waves with the same amplitude and wavelength and that travel together, find the displacement equation for the resultant wave and calculate the amplitude in terms of the individual wave amplitude and the phase difference.
- Describe how the phase difference between two transverse waves (with the same amplitude and wavelength) can result in fully constructive interference, fully destructive interference, and intermediate interference.
- With the phase difference between two interfering waves expressed in terms of wavelengths, quickly determine the type of interference the waves have.

Principle of Superposition of Waves

Let $y_1(x, t)$ and $y_2(x, t)$ be the displacements that the string would experience if each wave traveled alone. The displacement of the string when the waves overlap is then the algebraic sum

$$y'(x, t) = y_1(x, t) + y_2(x, t).$$

This summation of displacements along the string means that:

- Overlapping waves algebraically add to produce a **resultant wave** (or **net wave**).
- Overlapping waves do not in any way alter the travel of each other.

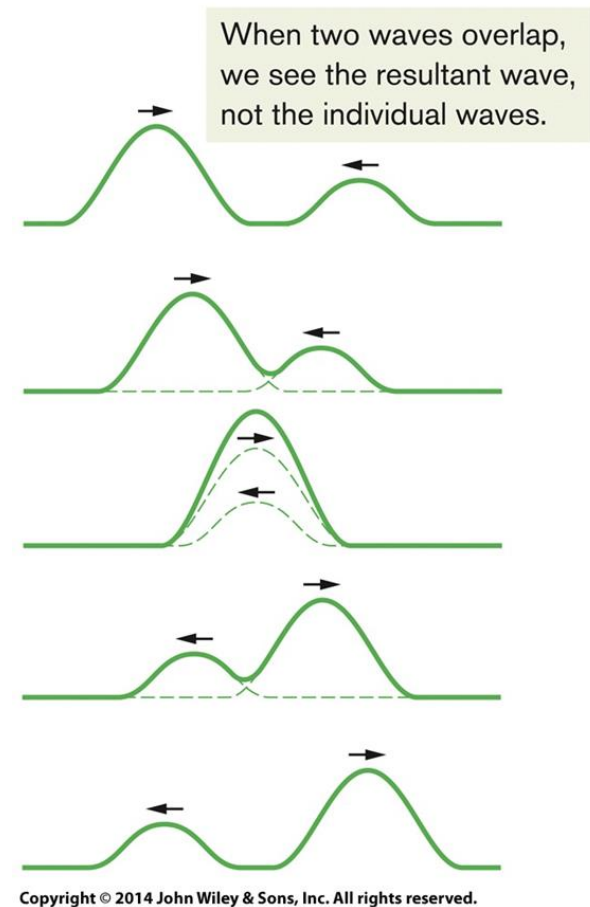


Figure 16.5.1

Displacement Expression for Interfering Waves

The resultant wave due to the interference of two sinusoidal transverse waves, is also a sinusoidal transverse wave, with an amplitude and an oscillating term.

$$\overbrace{y'(x,t)}^{\text{Displacement}} = \underbrace{[2y_m \cos \frac{1}{2}\phi]}_{\substack{\text{Magnitude} \\ \text{gives} \\ \text{amplitude}}} \underbrace{\sin(kx - \omega t + \frac{1}{2}\phi)}_{\substack{\text{Oscillating} \\ \text{term}}}$$

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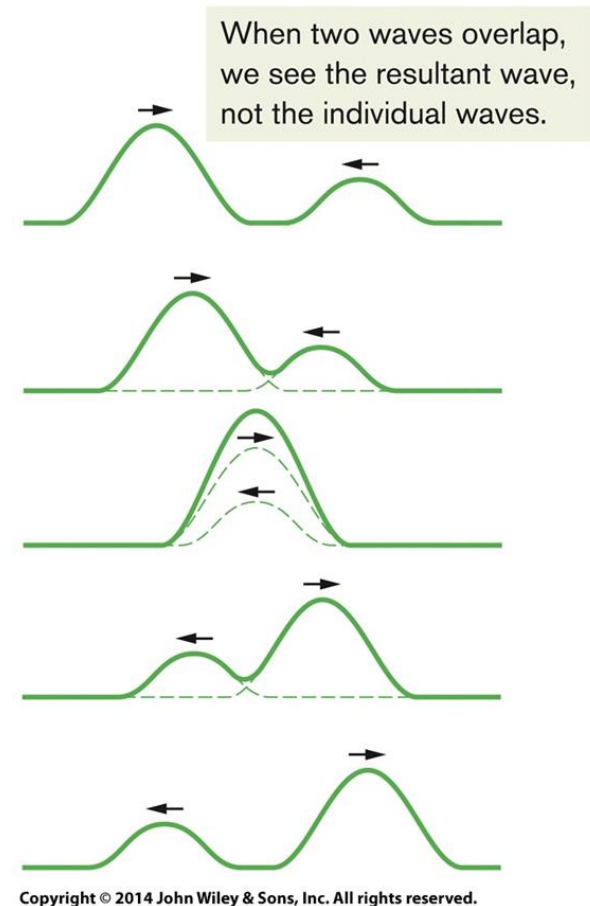
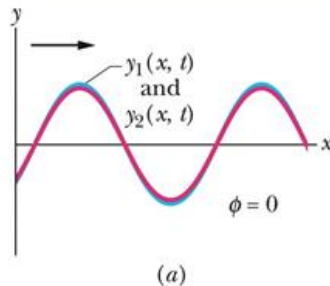


Figure 16.5.1

Constructive and Destructive Interference

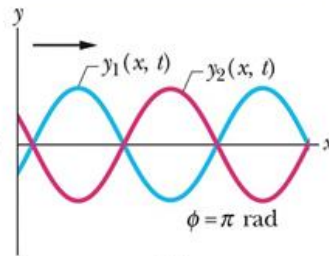
$$y'(x, t) = y_1(x, t) + y_2(x, t).$$

Being exactly in phase, the waves produce a large resultant wave.



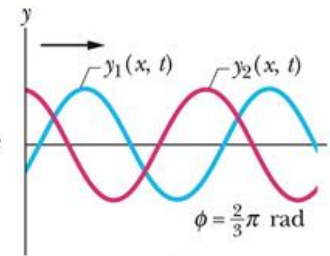
(a)

Being exactly out of phase, they produce a flat string.

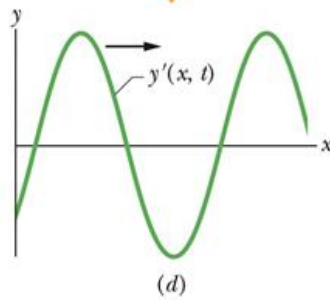


(b)

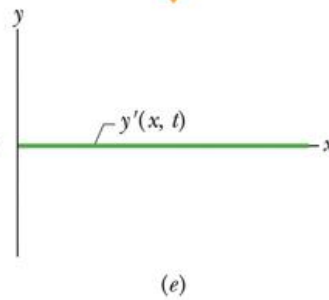
This is an intermediate situation, with an intermediate result.



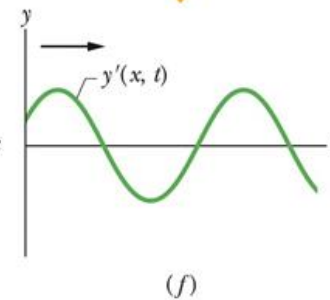
(c)



(d)



(e)



(f)

Figure 16.5.3

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Phase Dependence of Wave Interference

Two identical sinusoidal waves, $y_1(x, t)$ and $y_2(x, t)$, travel along a string in the positive direction of an x axis (Fig 16.5.3). They interfere to give a resultant wave $y'(x, t)$. The resultant wave is what is seen on the string. The phase difference ϕ between the two interfering waves is (a) 0 rad or 0° , (b) π rad or 180° and (c) $\frac{2}{3}\pi$ rad or 120° . The corresponding resultant waves are shown in (d), (e), and (f).

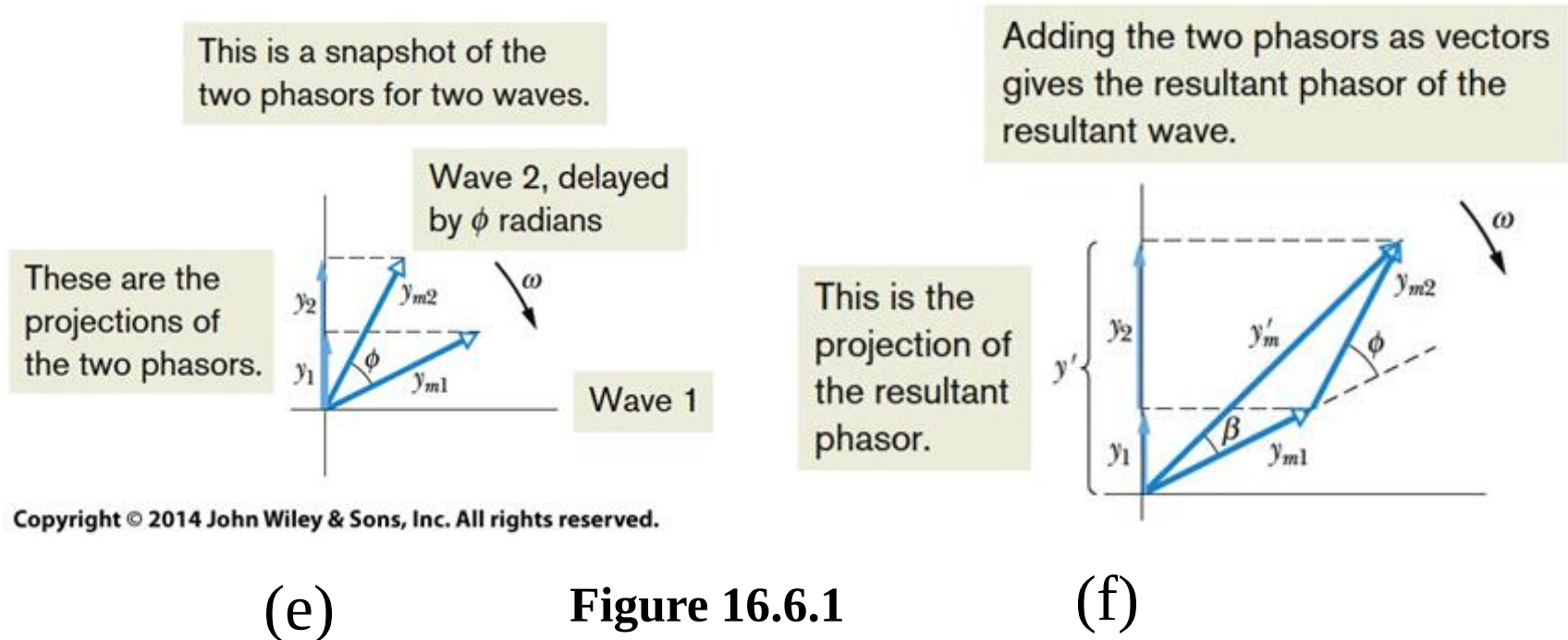
Section 16.6 Phasors

Phasors Learning Objectives

- Using sketches, explain how a phasor can represent the oscillations of a string element as a wave travels through its location.
- Sketch a phasor diagram for two overlapping waves traveling together on a string, indicating their amplitudes and phase difference on the sketch.
- By using phasors, find the resultant wave of two transverse waves traveling together along a string, calculating the amplitude and phase and writing out the displacement equation, and then displaying all three phasors in a phasor diagram that shows the amplitudes, the leading or lagging, and the relative phases.

Phasor as a Rotating Vector

A phasor is a vector that rotates around its tail, which is pivoted at the origin of a coordinate system. The magnitude of the vector is equal to the amplitude y_m of the wave that it represents.



Phasor Vector Addition to Represent Wave Interference

- e) A second phasor, also of angular speed ω but of magnitude y_{m2} and rotating at a constant angle β from the first phasor, represents a second wave, with a phase constant ϕ . (f) The resultant wave is represented by the vector sum $y' m$ of the two phasors.

Section 16.7 Standing Waves & Resonance

Standing Waves and Resonance

Learning Objectives #1 - #3

1. For two overlapping waves (same amplitude and wavelength) that are traveling in opposite directions, sketch snapshots of the resultant wave, indicating nodes and antinodes.
2. For two overlapping waves (same amplitude and wavelength) that are traveling in opposite directions, find the displacement equation for the resultant wave and calculate the amplitude in terms of the individual wave amplitude.
3. Describe the SHM of a string element at an antinode of a standing wave.

Standing Waves and Resonance

Learning Objectives #4 - #8

4. For a string element at an antinode of a standing wave, write equations for the displacement, transverse velocity, and transverse acceleration as functions of time.
5. Distinguish between “hard” and “soft” reflections of string waves at a boundary.
6. Describe resonance on a string tied taut between two supports, and sketch the first several standing wave patterns, indicating nodes and antinodes.
7. In terms of string length, determine the wavelengths required for the first several harmonics on a string under tension.
8. For any given harmonic, apply the relationship between frequency, wave speed, and string length.

Standing Waves

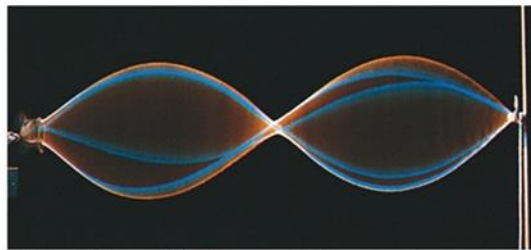
The interference of two identical sinusoidal waves moving in opposite directions produces standing waves. For a string with fixed ends, the standing wave is given by:

Displacement

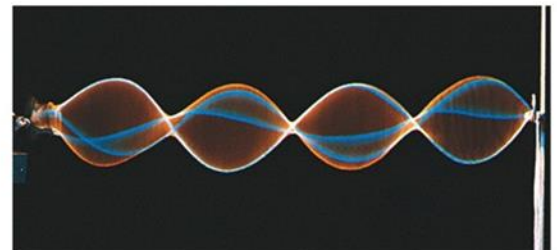
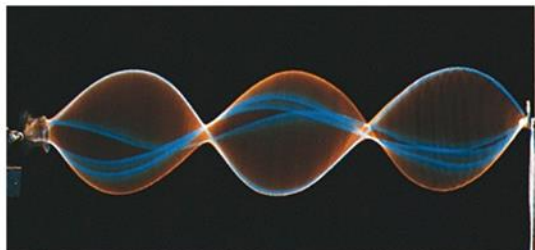
$$y'(x,t) = [2y_m \sin kx] \cos \omega t$$

Magnitude gives amplitude at position x Oscillating term

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Richard Megna/Fundamental Photographs



Harmonics

Standing waves on a string can be set up by reflection of traveling waves from the ends of the string. If an end is fixed, it must be the position of a node. This limits the frequencies at which standing waves will occur on a given string. Each possible frequency is a **resonant frequency**, and the corresponding standing wave pattern is an oscillation mode. For a stretched string of length L with fixed ends, the resonant frequencies are:

$$f = \frac{v}{\lambda} = n \frac{v}{2L}, \quad \text{for } n = 1, 2, 3, \dots$$

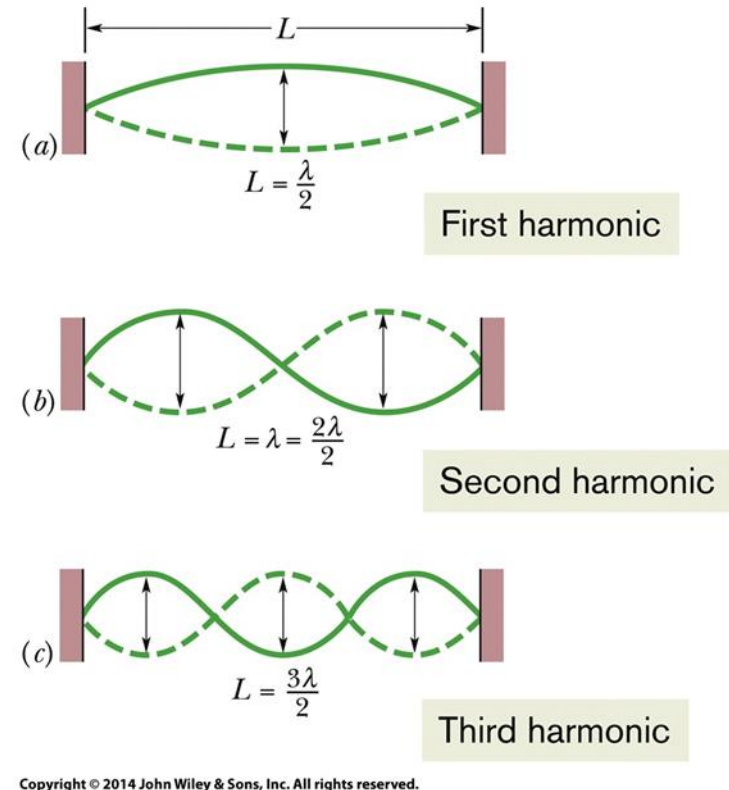


Figure 16.7.5

Chapter 16 Summary: Waves and Wave Speed

Waves

- Transverse Waves
- Longitudinal Waves

Wave Speed

- $\frac{\text{Angular velocity}}{\text{Angular wave number}}$

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f. \quad \text{Equation (16.1.13)}$$

Chapter 16 Summary: Sinusoidal and Traveling Waves

Sinusoidal Waves

- Wave moving in positive direction (vector)

$$y(x, t) = y_m \sin(kx - \omega t) \quad \text{Equation (16.1.2)}$$

Traveling Waves

- A functional form for traveling waves

$$y(x, t) = h(kx \pm \omega t) \quad \text{Equation (16.1.17)}$$

Chapter 16 Summary: Wave Power and Standing Waves

Wave Power

- Average Power is given by

$$P_{\text{avg}} = \frac{1}{2} \mu v \omega^2 y_m^2 \quad \text{Equation (16.3.7)}$$

Standing Waves

- The interference of two identical sinusoidal waves moving in opposite directions produces standing waves.

$$y'(x, t) = [2y_m \sin kx] \cos \omega t. \quad \text{Equation (16.7.3)}$$

Chapter 16 Summary: Wave Interference and Resonance

Interference of Waves

- Two sinusoidal waves on the same string exhibit interference

$$y'(x, t) = \left[2y_m \cos \frac{1}{2} \phi \right] \sin \left(kx - \omega t + \frac{1}{2} \phi \right). \quad \text{Equation (16.5.6)}$$

Resonance

- For a stretched string of length L with fixed ends, the resonant frequencies are

$$f = \frac{v}{\lambda} = n \frac{v}{2L}, \quad \text{for } n = 1, 2, 3, \dots \quad \text{Equation (16.7.9)}$$

Biomedical Applications: Problem 2

A human wave. During sporting events within large, densely packed stadiums, spectators will send a wave (or pulse) around the stadium (Fig. 16.6). As the wave reaches a group of spectators, they stand with a cheer and then sit. At any instant, the width w of the wave is the distance from the leading edge (people are just about to stand) to the trailing edge (people have just sat down). Suppose a human wave travels a distance of 853 seats around a stadium in 39 s, with spectators requiring about 1.8 s to respond to the wave's passage by standing and then sitting. What are (a) the wave speed v (in seats per second) and (b) width w (in number of seats)?

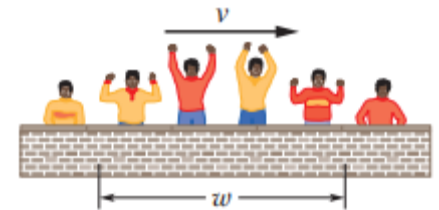


Figure 16.6

Answers: (a) 22 seats/s, (b) 39 seats

Biomedical Applications: Problem 4

A sand scorpion can detect the motion of a nearby beetle (its prey) by the waves the motion sends along the sand surface (Fig. 16.7). The waves are of two types: transverse waves traveling at $v_t = 50$ m/s and longitudinal waves traveling at $v_l = 150$ m/s. If a sudden motion sends out such waves, a scorpion can tell the distance of the beetle from the difference Δt in the arrival times of the waves at its leg nearest the beetle. If $\Delta t = 4.0$ ms, what is the beetle's distance?

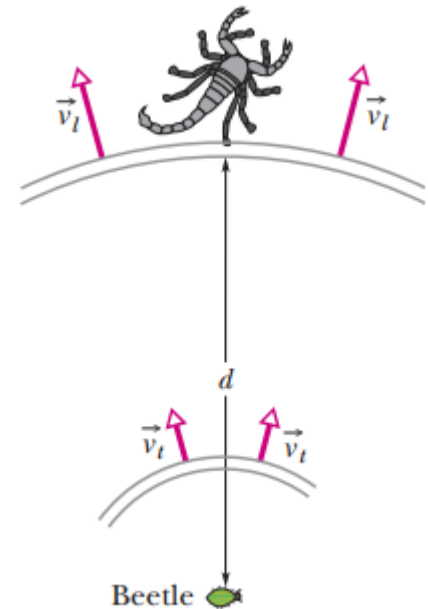


Figure 16.7

Answer: 30 cm